

Nikhil Kudache

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EDUCATION

University of Southern California

Master of Science in Computer Science

• Relevant Coursework: Analysis of Algorithms, Database Systems, Machine Learning for Data Science, Applied Natural Language Processing

Walchand Institute of Technology

Bachelor of Technology in Computer Science (GPA: 3.94/4.0)

• Relevant Coursework: Data Structures & Algorithms, OS, DBMS, OOP (Java/C++), Software Engineering, Cloud Computing, AI, Big Data.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, Bash

Web & Backend: REST APIs, Node.js, Next.js, React.js, PostgreSQL, MySQL, MongoDB

Cloud & DevOps: AWS (EC2, S3, ECS, Lambda), Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Linux/Unix

AI & Tools: Scikit-learn, Pandas, Git, Jira

EXPERIENCE

University of Southern California – Human Technology Interaction Lab

Software Engineer

- Developed secure RESTful API endpoints and JWT-based authentication for Phase 1 of a J&J MedTech surgical outcome digital twin platform used by 50+ surgeons, with server-side SAS token generation providing time-limited delegated access to sensitive medical data on Azure Blob Storage.
- Deployed the platform on Azure App Service with a GitHub Actions CI/CD pipeline, enforcing RBAC-only storage access via Managed Identity so no medical data was ever publicly exposed.
- Designed the data interface and backend architecture for a Verizon-funded educational platform supporting cross-functional product innovation, powering multimodal documentation and computational analysis of transdisciplinary learning outcomes.

Deloitte USI

Analyst – Cloud Infrastructure

- Engineered a Python automation tool that connected Prisma Cloud and Jira to detect when security alerts had been resolved in Prisma Cloud and auto-close the corresponding stale Jira tickets, clearing 300+ backlogged items in one run and saving the team 20 hours per week.
- Developed reusable Terraform modules for AWS networking, permissions, and compute resources, aligning deployments with Deloitte internal cloud security framework and client requirements based on NIST and CIS benchmarks, eliminating manual configuration errors across teams.
- Integrated Checkov and Prisma Cloud into a dedicated security scanning stage of the Jenkins CI/CD pipeline that ran on every push, blocking 90+ high-severity vulnerabilities per release cycle from reaching production.

TECHNICAL PROJECTS

Evaluating Reputation Effects in Multi-Issue LLM Negotiation | Python, Llama 3.1

Mar 2026 – Present

- Designed a multi-agent LLM negotiation framework using Llama 3.1 via Ollama, where buyer and seller agents bargain over price, delivery time, and warranty through structured Thought-Talk-Action prompt patterns across 100+ episodes, supporting multiple inference backends (Ollama, Groq, Anthropic).
- Measured reputation visibility improved deal rate by 6% and buyer utility by 6% over a no-reputation baseline, documenting findings for knowledge sharing.

Talent Management System (AI Academy) | Python, PySpark, SQL, Scikit-learn, Azure, Power BI

Jan 2023 – Mar 2023

- Trained a machine learning model that predicted candidate drop-off during onboarding with 85% accuracy, helping HR teams intervene early and reduce attrition by up to 20%.
- Built scalable data pipelines with PySpark and SQL to process large recruitment datasets for model training.
- Built real-time analytics dashboards in Power BI on Azure, giving stakeholders a live view of recruitment trends and model predictions.

Nagar and Associates | React.js, GatsbyJS, Tailwind CSS, Netlify CMS, Google Analytics

May 2021 – Jun 2021

- Developed a responsive client-facing website for a tax consultancy firm in React.js, GatsbyJS, and Tailwind CSS, enabling non-technical staff to update service listings and content through a CMS without touching code.
- Integrated Netlify CMS for non-technical content management, custom email workflows via Netlify Forms, and Google Analytics for traffic analysis and user behavior tracking.

Private Cloud Infrastructure | MinIO, Docker, Portainer, Cloudflare, Tailscale, Raspberry Pi 5

Jan 2026 – Present

- Built a self-hosted S3-compatible storage system on a Raspberry Pi 5 with MinIO and Docker, managing the full infrastructure lifecycle end-to-end as a personal project.
- Deployed secure remote access for the home lab via Cloudflare Tunnels and Tailscale, keeping zero ports exposed to the public internet and routing traffic through encrypted tunnels between nodes in two countries.

AWARDS & RECOGNITION

Applause Award – Deloitte

Apr 2025

- Recognized for building the automation system that eliminated 30% of manual task assignments and saved the cyber infrastructure team 20 hours per week.